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Universal Patterns in Eye Movements Across Lines of Text

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This study examines cross-linguistic patterns of eye movements before, during, and after return sweeps—the saccades that move readers' gaze from one line of text to the next. Using data from 28 language samples representing 22 languages and multiple writing systems from the Multilingual Eye Movement Corpus project, we tested whether spatial and temporal characteristics of interline transitions reflect universal oculomotor mechanisms or language-specific factors. Results show that extreme fixations cover about 80% of each line across all languages, and fixation durations follow consistent positional patterns: shorter at line ends and longer at line beginnings. Neither linguistic nor script properties, nor reading direction (horizontal vs. vertical), significantly modulated these effects. These findings suggest that return-sweep patterns are highly stereotypical across writing systems and primarily determined by visuo-oculomotor rather than linguistic processes, representing a potential universal of human reading behavior.

Public Significance Statement

Reading involves a complex coordination of visual, oculomotor, and linguistic processes. Eye movements that bring the gaze from the end of one line of text to the beginning of the next—return sweeps—have only been studied in a handful of languages so far. This study asks whether the eye-movement behavior around line transition is driven by linguistic factors (and thus specific to individual languages) or is universal (and thus driven by visual and oculomotor factors). This study represents a substantial theory-driven advance because it examines interline transitions across 22 languages and four writing systems, using nearly three million fixations from the Multilingual Eye-Movement Corpus. We found that the spatial and temporal characteristics of return sweeps were remarkably uniform across languages and reading directions, including horizontally and vertically written scripts. This work identifies return sweeps as a universal component of skilled reading and advances our understanding of how visual and cognitive systems jointly coordinate the act of reading across human languages. It also informs computational models of reading.

Keywords: eye movements, visual attention, cross-linguistic comparisons, behavioral universals

Supplemental materials: <https://doi.org/10.1037/xhp0001419.supp>

Return sweeps in reading are eye movements that bring the eye gaze from one line of text to the next. These saccades are

regressive, i.e., go against the direction of reading, and have much longer amplitudes and durations than the saccades that move the reader's gaze within the line of text. The behavioral phenomenon of return sweeps as well as fixations that precede and follow them have piqued the researchers' interest for well over a century (Dearborn, 1906; Paterson & Tinker, 1940; Rayner, 1978). This is because gaze transitions between the text lines provide a special window into the inquiry into the division of labor between oculomotor control, visual processing, and linguistic processing. Yet between-line transitions remain understudied compared with other types of eye movements in reading, partly because the most common stimulus in this research field is a single sentence, i.e., the one-line visual object that does not elicit return sweeps (Parker et al., 2025). Moreover, to our knowledge, the extant studies of return sweeps and related eye movements have been primarily conducted in English, with only a few investigations in German and Chinese (see reviews by Hofmeister et al., 1999; Slattery & Parker, 2019; Sun et al., 1985; Wang et al., 2024). This article aims to overcome this linguistic bias and considers return sweeps

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from a cross-linguistic perspective by analyzing text reading data from 28 samples of 22 languages, representing a broad variety of writing systems. The remainder of the Introduction briefly summarizes the amassed knowledge about interline gaze transitions, indicates areas with incomplete or lacking evidence, and formulates research questions and approaches of the present investigation.

Interline Gaze Transitions

The spatial and temporal characteristics of the gaze's transition from the end of one line to another line are robust and well-described in children and adults (Parker et al., 2019). Readers tend to launch the return sweep saccade without fixating the last many characters of the line: This rules out the readers' ability to have a visual uptake of the remaining characters of the launch line foveally and sometimes even parafoveally (Hofmeister et al., 1999). Furthermore, the return sweep saccade tends to undershoot the beginning of the next line, landing several characters away from the text boundary. The undershoot arguably stems for the saccadic range error that is proportional to the saccade amplitude and leads to a lower precision of saccadic targeting after longer saccades (e.g., McConkie et al., 1988). Since the launch site and the landing site of the return sweeps are too remote to allow the parafoveal preview of the upcoming line, readers do not identify or target the center of the line-initial word as the optimal viewing position, and instead land further into the line (Hofmeister et al., 1999; Slattery & Parker, 2019; Slattery & Vasilev, 2019). The inability to preview the beginning of the next line is also confirmed by the observation that the length and other lexical characteristics of the line-initial words do not influence the landing position of the return sweeps (Hofmeister et al., 1999; Parker et al., 2019; Radach & Kempe, 1993). Because they commonly undershoot the line-initial characters, return sweeps are often followed by a corrective regression that brings the gaze closer to the beginning of the line: Slattery and Vasilev (2019) estimate this behavior to characterize 40%–60% of return sweeps in English text reading. As a result of the two tendencies—to avoid fixating last characters of one line and initial characters of the next line—Rayner's review (1998, p. 375) estimates that only about 80% of the text falls between extreme fixations in the line.

Besides these spatial characteristics, transitions across lines have a distinct temporal profile. Fixations that occur between the return sweep and the corrective regression to the beginning of the line, labeled “undersweeps” in the literature (e.g., Parker et al., 2017), tend to be very short (estimated at 128–156 ms in the study of English text reading by Parker et al., 2020) and do not demonstrate influence of linguistic properties of the fixated word. However, line-initial fixations that are followed by the forward saccade and initiate the regular program of reading a line are longer than the intraline fixations. As stated in Rayner's review (1978: 621), “this could be due to the possibility that the reader has not had the opportunity to begin a grouped program of left-to-right saccades on the first fixation after the return sweep.” Finally, the last fixation on the line, which precedes the return sweep, is usually shorter than the intraline fixations. Arguably, this is because its primary aim is planning and launching the return sweep to the next line rather than allocating attention to the final characters of the currently fixated line.

The specific causes of the characteristic oculomotor behavior accompanying return sweeps are still debated. Although there is no doubt that oculomotor coordination and visual processing code-terminate both the spatial and temporal parameters of line transition, the role of linguistic processing is more uncertain, see above. Parker et al. (2025) compared return sweeps in the regular English text reading and “mindless reading,” i.e., scanning of the strings consisting of z-characters that were length-matched with the texts but were otherwise void of linguistic material. Since both tasks demonstrated shorter fixations at the line endings, Parker et al. (2025) attributed this reduced fixation duration to oculomotor coordination or visual processing, rather than linguistic processing. Although both text reading and z-string scanning showed a reliable difference between accurate line-initial and intraline fixations, the difference was greater in text reading.

What Can Cross-Linguistic Data Reveal?

Although highly valuable, existing research on interline gaze transitions is based on a very small set of languages, reflecting the broader bias of eye-tracking studies toward alphabetic European languages and Chinese (Angele & Duñabeitia, 2024; Siegelman et al., 2022). Moreover, perhaps the most incisive current evidence comes from the comparison of text reading against nonlinguistic tasks with low ecological validity and very different attentional and cognitive demands (Parker et al., 2025; Schad & Engbert, 2012). We argue that relying on cross-linguistic data can shed light on the long-standing question of whether return-sweep patterns are primarily shaped by oculomotor, visual, or linguistic factors or a combination of these. Written languages of the world vary dramatically in their visual, orthographic, phonological, morphological, and syntactic properties. Writing systems also differ markedly in the physical size and informational density of their orthographic units: Logographic scripts like Chinese encode more information per character than alphabetic systems, while scripts such as Hangul, Mongolian, or Hindi structure characters into syllabic or abugidic blocks with high information density per unit of space (Huang et al., 2024). Thus, if a behavioral phenomenon is observed with a degree of uniformity across multiple written languages and especially writing systems, it is very likely to be of oculomotor or low-level visual nature, relying on the physiological, neurological, and visuo-perceptual substrates that are shared by humans, and thus readers of any language. Substantial cross-linguistic variability in that behavior would signal a likely linguistic or experiential origin, which may be attributable to the specific properties of the written languages under study. This theorizing is at the core of the research program that aims to determine universal versus specific features of reading behavior across multiple languages (Kuperman et al., 2024; Li et al., 2022; Liversedge et al., 2016; Siegelman et al., 2022).

Application of this theoretical framework to the present study yields the following space of possibilities. If linguistic properties do influence return-sweep planning and execution, we should expect cross-linguistic variability in the distance covered between extreme fixations. Since languages vary widely in the information content associated with a character (Huang et al., 2024), omitting the same physical space or the same number of characters between fixations may have different effects on reading efficiency across languages. Therefore, if return sweeps are dependent on linguistic

factors, scripts with dense informational content should show extreme fixations closer to text edges (see Huang et al., 2024). If linguistic factors also codetermine durations of either line-final or line-initial fixations, we would expect appreciable cross-linguistic variability in those durations. This is because languages differ in multiple structural and lexical properties, such as average word length (e.g., short in English, long in Finnish) and average relative word frequency (lower in Finnish than English because of the rich morphological system generating multiple inflectional forms of the same root word). In contrast, if return sweeps are primarily visually driven, such linguistic differences should exert minimal influence on fixation timing.

Another reason to consider cross-linguistic evidence is that oculomotor behavior during line transitions may vary as a function of reading direction. Although a return sweep in the left-to-right writing systems (like the Roman script that English uses) takes the gaze from right to left and downward, the same eye movement in the right-to-left systems (used in Arabic and Hebrew) moves the gaze from left to right and downward. In the vertical script of traditional Mongolian, which is written from top to bottom and left to right, the return sweep goes from the bottom of the page to the top and rightward. The visual and oculomotor systems exhibit a well-established horizontal-vertical anisotropy, that is, the visual performance field is broader and performance in perceptual and oculomotor tasks is consistently superior along the horizontal meridian than along the vertical one (Fuller et al., 2008; Greenwood et al., 2017; Himmelberg et al., 2020). This asymmetry reflects multiple underlying factors, including distinct neural pathways responsible for generating horizontal and vertical saccades, as well as anatomical asymmetries in the eyes and visual field that afford a wider range of motion and superior perceptual sensitivity along the horizontal meridian. Humans also engage in predominantly horizontal eye movements in everyday vision to maintain binocular alignment and depth perception (Liang et al., 2005). Consequently, visual processing along the vertical meridian tends to be slower and less accurate than along the horizontal meridian in basic perceptual tasks. Extending this evidence to natural reading, Bao, Yan, et al. (2025) showed that vertical saccades are slower and reach lower average and peak velocities than amplitude-matched horizontal saccades during natural passage reading. Thus, it is possible that reading in vertical scripts, including traditional Mongolian that is part of Multilingual Eye-Movement Corpus (MECO), alters spatial or temporal characteristics of return sweeps and fixations that frame those.

Thus, the present article tests robustness of the widely reported eye movement patterns around line transitions when considered against multiple written languages and writing systems. The specific patterns we target are as follows: (a) the first and the last fixations landing on a line only cover about 80% of the line length and neglect the visual inspection of the line-initial and line-final characters; (b) line-initial fixations are often followed by a corrective regression bringing the gaze closer to the beginning of the line; (c) line-initial fixations are longer than intraline fixations; and (d) line-final fixations are shorter than intraline fixations. The first research question (*RQ1*) is whether behavioral patterns (a)–(d) do give rise to appreciable cross-linguistic variability, and if they do, whether this variability is random or can be systematically attributed to the specific properties of those languages. *RQ1* is grounded in the programmatic cross-linguistic research that aims

at determining universal versus specific aspects of reading behavior (Li et al., 2022; Liversedge et al., 2016). The space of logical possibilities that this research defines is the following. Observing variability in behavioral patterns that is minimal and practically unimportant would provide strong evidence in favor of universality of behavioral patterns across languages and writing systems. Given the diversity of languages, minimal variability would also suggest that linguistic factors do not contribute to the behavioral patterns, and those are rather driven by oculomotor or visual factors shared by readers of any language. An alternative possibility is that we will observe substantial variability across languages in the studied behaviors. If this variability can be linked to structural or formal properties of the languages, the patterns can be regarded as language-specific, and thus linguistic factors are at play. A final possibility is that behavioral patterns are expressed randomly across languages, in which case they likely do not constitute a foundational component of oculomotor behavior during reading.

The second research question (*RQ2*) is whether the direction of reading influences oculomotor behavior during and around gaze transition across lines. To this end, we compare patterns recorded in the vertically read traditional Mongolian to those found in horizontal languages. The criterion for determining the presence of a reading direction effect is whether the oculomotor patterns observed in traditional Mongolian reading stand out in their spatial or temporal parameters from those of a broad range of horizontally read languages.

The discussion hereafter is organized in two studies. Study 1 addresses the spatial characteristics of return sweeps and the locations of extreme fixations on the line and their coverage of the available text, point (a) above. Study 2 focuses on the cross-linguistic comparison of fixation durations as a function of their location in the line, points (b)–(d). Both studies touch upon the two research questions of this article. We preface the reporting of the studies with a brief description of the data set and data preprocessing.

Method

The data sets used in this secondary reanalysis include a total of 28 language samples published as Wave 1 (Siegelman et al., 2022) and Wave 2 (Siegelman et al., 2025) of the MECO project, as well as the data set in traditional Mongolian (Bao, Li, et al., 2025). Data collection for all language samples followed the same experimental protocol and procedure, including the use of EyeLink eye-trackers (1000, 1000+, or Duo). There were unavoidable minor variations in the characteristics of the screens (size, resolution) used in different laboratories, the reader's distance from the screen, as well as font type and size, spacing, and text margins, as required by the visual form of the specific written language and the experimental setup. Since our mixed-effects regression models, see hereafter, code each language sample separately (as a combination of the language and site of data collection), this variability is likely absorbed by Language as a fixed effect.

The texts were left-aligned for all languages written in the left-to-right direction, right-aligned for Hebrew (written right-to-left), and top-aligned for traditional Mongolian (written top-to-bottom left-to-right). In all but two cases, monospaced fonts were used, in which each character occupies the same number of pixels. The

exceptions were Traditional Mongolian and Serbian. Traditional Mongolian is not written in separable single letters. Instead, consonants and vowels combine into syllabic units, and each syllable changes its shape depending on its position within a word—initial, medial, or final. As a result, syllables vary in visual length and width, making it impossible to represent the script accurately with a monospaced font (see example illustration in Yan et al., 2025). Serbian used a proportional font due to the programming error. Information about the technical characteristics of the setup at each testing site is compiled from the respective published studies and provided in the Supplementary materials S1. Participants were first-language speakers of the language of testing and (mostly undergraduate) university students and staff members. Detailed information about participants, their age distribution, and compensation is compiled from published reports and provided in the Supplementary materials S2. All paragraph reading data, except Traditional Mongolian, was processed for automatic correction of fixation locations using the popEye software (Schroeder, 2019). The popEye software is an integrated environment to preprocess and analyze eye-tracking data from reading experiments and assigns fixations to lines, words, and letters: see Supplementary materials to Siegelman et al. (2022, 2025) for further details. Fixation alignment with the vertical script of Traditional Mongolian was checked manually (see Bao, Li, et al., 2025).

Transparency and Openness

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study, and we follow JARS (Appelbaum et al., 2018). All data, analysis code, and research materials are available at https://osf.io/nsr45/overview?view_only=439c4eeefaa4f99b18722dc08802512 (Kuperman & Bao, 2025). Data were analyzed using R, Version 4.5.0 (R Core Team, 2025) and the package ggplot2, Version 3.5.2 (Wickham, 2011). This study's design and its analysis were not preregistered. This study analyzes previously published corpus data available in open access and does not require ethics clearance.

Data Analysis

We made use of fixation reports for these language samples, where information is organized per fixation, including its location in the line of text and duration. The total data set consisted of 2,884,719 fixations.

Study 1

For the exploration of fixation locations around return sweeps in Study 1, we identified locations of the fixations that were the first and last to land on every line (i.e., the line-initial and -final fixations). The y-coordinate of the location was used for traditional Mongolian (written top-to-bottom), and the x-coordinate in all other languages. We also identified the positions of the first and last character on each line, as the naturally constrained spatial boundaries for the fixations on the text. For comparability, in the further analyses, all locations are considered in pixels rather than letters, and relative to the beginning and the end of the line in the language's direction of reading. This data pool of line-level information consisted of 147,776 observations.

The trimming procedures were as follows. Final lines of each text were removed from consideration (13,173, .09%), because they often occupied a smaller portion of the screen than the nonfinal lines. A small number of observations (<.5% of the total data set) had the initial or final fixations recorded outside of the line boundaries, typically, within 5 pixels, either due to imperfections of the eye-tracking recording or the rounding error. Such observations were removed from consideration. Furthermore, we removed lines in which locations of the line-initial or -final fixations did not conform with the regular reading behavior for comprehension. These were cases where the first fixation on the line landed on the last third of the line length, or the last fixation landed on the first third of the line length. After this trimming, the data set for Study 1 consisted of 109,211 pairs of the first and last fixations and their locations.

The next step was to fit the linear mixed-effects multiple regression models to the relative locations of the line-initial and -final fixations, separately. To this end, all locations were recalculated as relative locations, i.e., as a percent of line length, in pixels. The models fitted to these relative fixation locations included the intercept, the categorical variable Language with 28 levels, corresponding to the number of language samples, and random intercepts for participant and text. The model structure was as follows: $dv \sim 1 + \text{Language} + (1 | \text{Participant}) + (1 | \text{Text})$, where dv was relative locations of the line-initial or line-final fixations. Sum-coding was used for the contrasts of this categorical variable.

The following step was to formally determine the degree of cross-linguistic variability in the locations of extreme fixations. To this end, we fitted intercept-only models to line-initial and, separately, line-final fixations. These models had the following structure: $dv \sim 1 + (1 | \text{Participant}) + (1 | \text{Text})$, where dv stood for relative locations of the line-initial or line-final fixations. The comparisons of the two respective sets of regression models using the log-likelihood test determine whether cross-linguistic variability in fixation locations is statistically significant. Random by-text slopes of Language were attempted by led to convergence issues and thus were removed from all models.

Study 2

This section considers another aspect of the behavior that readers exhibit when transitioning from one line of text to the next one. As stated in the literature (e.g., Dearborn, 1906; Hawley et al., 1974; Parker et al., 2025; Rayner, 1978), fixation durations vary as a function of their position in the line. Line-final fixations, from which return sweeps are launched, are shorter than intraline fixations. Line-initial fixations are longer than intraline fixations (and the line-final fixations) if they land in a convenient position for reading and are followed by forward saccade that propels the gaze along the text line. However, most line-initial fixations (68% in our data) are followed by a corrective regression toward the line beginning, because the landing positions of such fixations tend to overshoot the first word(s) (Rayner, 1978). Such line-initial fixations are called “undersweeps” and tend to be very short, i.e., shorter than the intraline saccades. Throughout the literature, this variance in fixation durations is attributed to oculomotor mechanisms of planning, targeting, and execution of long-range saccade and subsequent word recognition, see the literature review in Slatery and Parker (2019) and Parker, Tao, and Vasilev (2025). We set out to test whether the same durational patterns hold across all

languages. If they are truly of oculomotor nature and thus independent of linguistic and visual properties of written materials, we expect to see uniform behavior in all writing systems and languages.

Similarly to the aforementioned section, we extracted durations for first and last fixations from each line (except for the last line in the text), and calculated the average fixation duration for intraline (not line-initial or -final) saccades. We further classified line-initial fixations into undersweeps (followed by a regression, i.e., a saccade against the direction of reading) and accurate fixations (followed by a forward saccade). The resulting data pool consisted of 42,411 accurate line-initial fixations, 84,181 undersweeps, 126,592 mean values for intraline fixations, and 126,592 line-final fixations. Given these substantial sample sizes, statistical power is sufficient and is not a limiting factor for the analyses reported hereafter.

As the next step, a linear mixed-effects model was fitted to fixation duration as the dependent variable, and an interaction between the categorical variable position (line-initial accurate, line-initial undersweep, intraline, or line-final fixation position) and the 28-level categorical variable Language as predictors. The model also included by-participant intercepts as a random effect. The model included no intercept and was used to estimate the fixation duration in each position and for each language. The model structure was as follows: fixation duration $\sim 0 + \text{Position} : \text{Language} + (1 | \text{Participant})$. Inclusion of the random intercepts by Text or of the random by-participant slopes of Position led to convergence issues: These random effects were not included in the final model.

Practical importance. Our interpretation of results below relies on the notions that statistical significance of findings may differ from its practical importance (see Gelman & Stern, 2006, for an excellent overview). Although the pitfalls of the (frequentist) statistical hypothesis testing are widely demonstrated (McShane et al., 2019), the criteria for what counts as practical importance are necessarily field-dependent and thus may be subject to debate. We based our definition of practical importance on the measurement accuracy of the eye-tracking devices. Spatial accuracy of the EyeLink devices used in this study is advertised by the manufacturer (SR Research) as .25°–.5° of the visual angle. Most samples of alphabetic languages in MECO reported one degree of the visual angle to subtend three to four characters, see the Supplementary materials S1. The maximum values from both estimates (.5° and four characters per 1°) yield two characters as the estimated measurement error for spatial accuracy. The average line length in the alphabetic languages using Roman scripts in MECO was 90 characters. Expressed as the percentage of line length, the spatial measurement error of the eye-tracking devices that we used was 2.2% (2/90 characters). In this study, estimated variability of eye movement measures that had a standard deviation below this measurement error was deemed practically unimportant.

Results

Study 1: Spatial Placement of Line-Initial and Line-Final Fixations

Across all 28 language samples, the reader's extreme fixations (i.e., the first and the last to land on the line) were estimated to

cover 81% (*SD* 3%) of the text, almost identical to Rayner's (1998) estimation of the 80%. The difference between relative locations of extreme fixations ranged from 73% for the Canadian English sample (labeled en) to 84% for the Serbian sample (se). The estimation across the entire data set shows the average relative location for line-initial fixations to be at 9% (*SD* 1%) and for line-final fixations at 90% (*SD* 2%) of the line length, as indicated by the intercepts for the respective models. Figure 1 visualizes the results of modeling for relative fixation locations. The estimated locations for extreme fixations and the text coverage between these locations are reported for each language in the Supplementary materials S3.

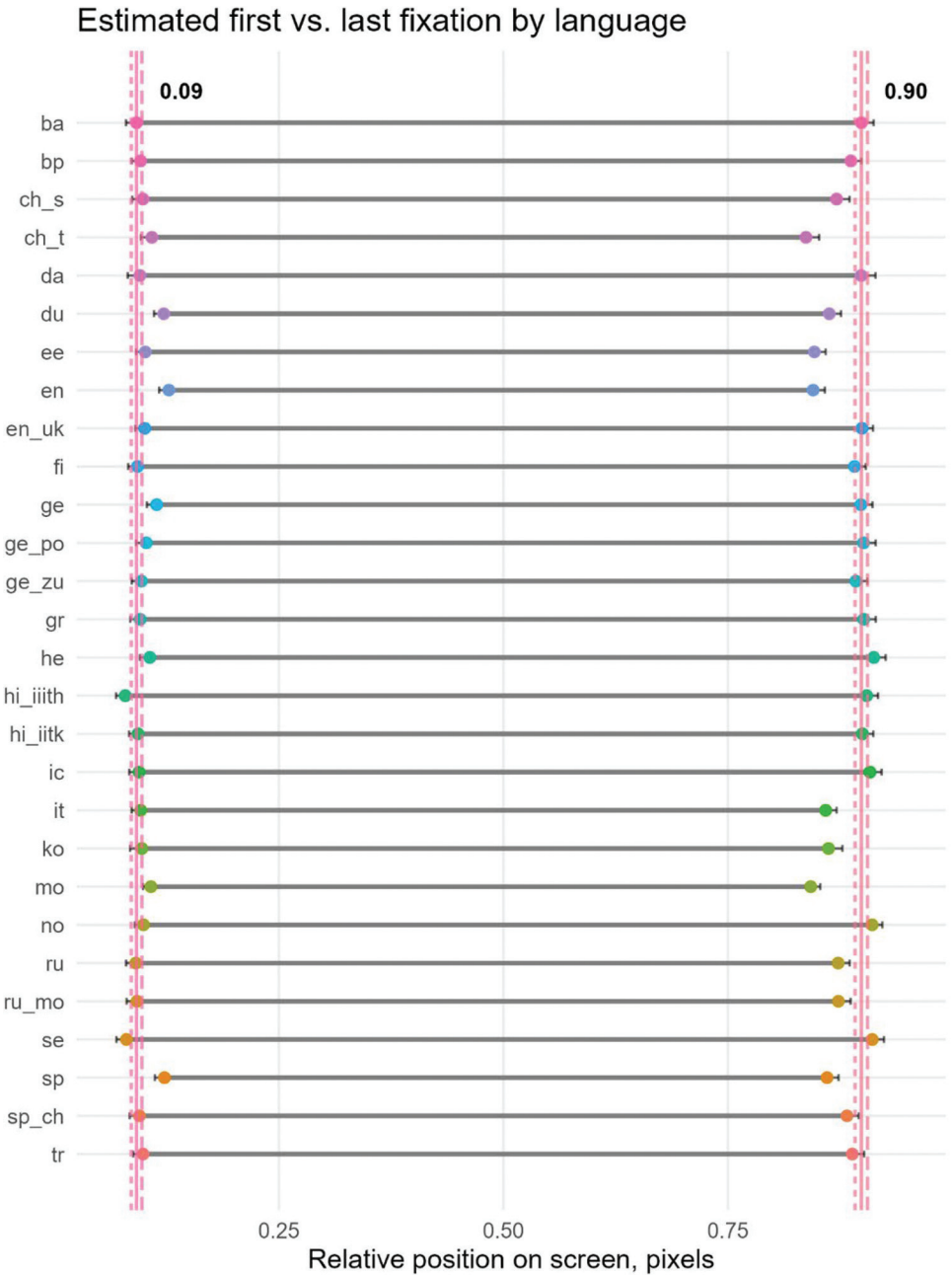
As the next step, we compared the full models (including Language) to the intercept-only models using the log-likelihood test. The full models with Language as a predictor provided a significantly better fit ($p < .001$) to the line-initial or -final fixation locations than the intercept-only models.

Although statistically significant, the practical importance of cross-linguistic variability appears to be quite minor. Locations of the first fixations on the line were only reliably different from the grand average in six language samples: Traditional Chinese (ch_t), Dutch (du), Canadian English (en), German (ge), traditional Mongolian (mo), and Argentine Spanish (sp). There was more spread in the locations of line-final fixations, but the standard deviations were small for both extremes (1% for line-initial and 2% for line-final fixations), i.e., there were below the spatial measurement error of the eye-tracking devices, see the Data Analysis section described earlier in the text.

Moreover, it appears unlikely that this variability was systematically related to linguistic properties of the written languages. Perhaps the strongest evidence against systematicity comes from the two English samples in the MECO data set, collected in Canada (en) and the U.K. (en_uk). As mentioned earlier, the Canadian sample demonstrated the smallest text coverage: Although the U.K. sample used the identical stimuli and presentation format, it showed a much broader text coverage than the Canadian counterpart, 80% versus 72% ($p < .001$ as indicated by the two-sample proportion test). We infer, in response to *RQ1*, that the observed cross-linguistic variability is not driven by the visual, typographic, or linguistic characteristics of the text. It may, in part, stem from differences of the experimental setup (including the distance from the screen and other screen characteristics), as well as the reader's motivation toward engaging in the reading for comprehension task.

The data patterns in Figure 1 provide answers to *RQ2* as well. The vertically read traditional Mongolian did not stand out from the rest of the samples. Reading in this language did not produce the minimum nor the maximum value for either the line-initial (11%) or final fixation position (84%). Although the amount of text coverage for traditional Mongolian readers was small (73.5%), it was only the third smallest among other languages represented in MECO (following the Canadian English 71.8% and the traditional Chinese samples 72.9%). That is, vertical saccades going upward and to the right from the end of one line of the traditional Mongolian text to the beginning of the next line have the same spatial characteristics as the long-range return sweep regressive saccades in horizontally written languages. Generally, we conclude that the direction of reading does not radically alter the launch sites and landing positions of the return sweep saccades.

Figure 1
Estimated Positions of the Line-Initial and Line-Final Fixations for Each Language, Relative to the Total Line Length in Pixels



Note. Error bars stand for 1 *SEM*. Vertical solid lines show the estimated grand average for line-initial and -final fixation positions; dashed lines mark the respective ± 1 *SEM*. Language codes correspond to each data set sample: ba = Basque; bp = Brazilian Portuguese; ch_s = Chinese (Mandarin, simplified); ch_t = Chinese (Mandarin, traditional); da = Danish; du = Dutch; ee = Estonian; en = English (Canada); en_uk = English (UK); fi = Finnish; ge = German; ge_po = German (Potsdam sample); ge_zu = German (Zurich sample); gr = Greek; he = Hebrew; hi_iitk = Hindi (Kanpur sample); hi_iith = Hindi (Hyderabad sample); ic = Icelandic; it = Italian; ko = Korean; mo = Traditional Mongolian; no = Norwegian; ru = Russian; ru_mo = Russian (Moscow sample); se = Serbian; sp = Spanish; sp_ch = Spanish (Chile sample); tr = Turkish. See the online article for the color version of this figure.

Besides this qualitative similarity, numeric estimates of extreme fixations and text coverage are highly similar across languages. This suggests a substantial uniformity in the spatial parameters of return sweeps across a wide variety of written languages and writing systems.

Study 2: Fixation Durations by Position

Our first target phenomenon was the likelihood of the line-initial fixation to be followed by a regression rather than begin a sequence of fixations in the direction of reading. Sixty-seven percent of line-initial fixations were followed by a corrective regression toward the line beginning, showing a full convergence between the cross-linguistic data set with the percentage (68%) reported for English text reading by Parker et al. (2025). This convergence suggests that undershooting the beginning of the line in a return sweep and subsequent moving gaze closer to the line beginning is a universal feature of reading behavior that holds across multiple written languages and writing system. As

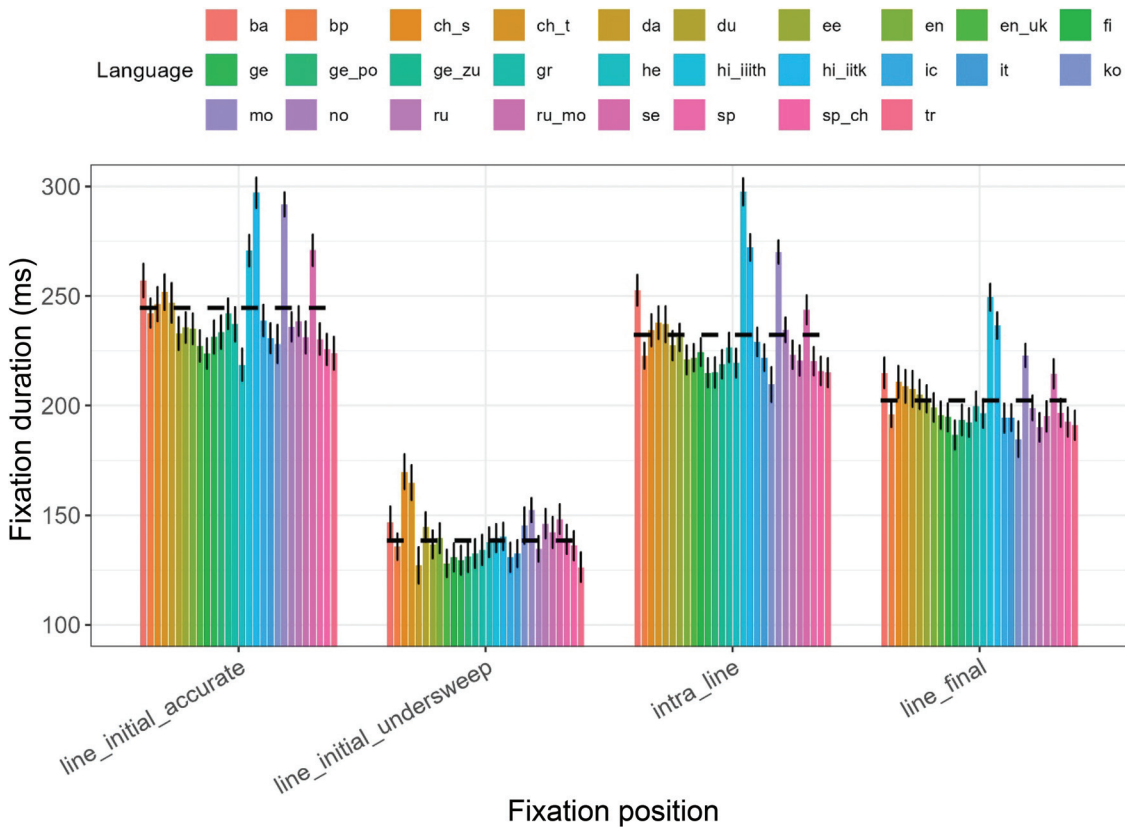
such, it is unlikely to be of a linguistic nature and rather is driven by oculomotor and visual factors.

As described in the Data Analysis section, a linear mixed-effects model was fitted to fixation duration as the dependent variable, and an interaction between the categorical variable Position (line-initial accurate, line-initial undersweep, intraline, or line-final fixation position) and the 28-level categorical variable Language as predictors. Estimated fixation durations for each position and language are reported in the Supplementary materials S4. Figure 2 illustrates the partial effects of the Position $\times$ Language interaction of the regression model. Figure 3 visualizes the numeric difference between the durations of the line-final fixation and the intraline fixation (left) and the durations of the accurate line-initial fixation and the intraline fixation (right), by language.

The patterns of fixations by position in the line appear highly stereotypical both across languages and within each language. We used estimated marginal means from the fitted regression models to compare numeric values of fixation durations across position

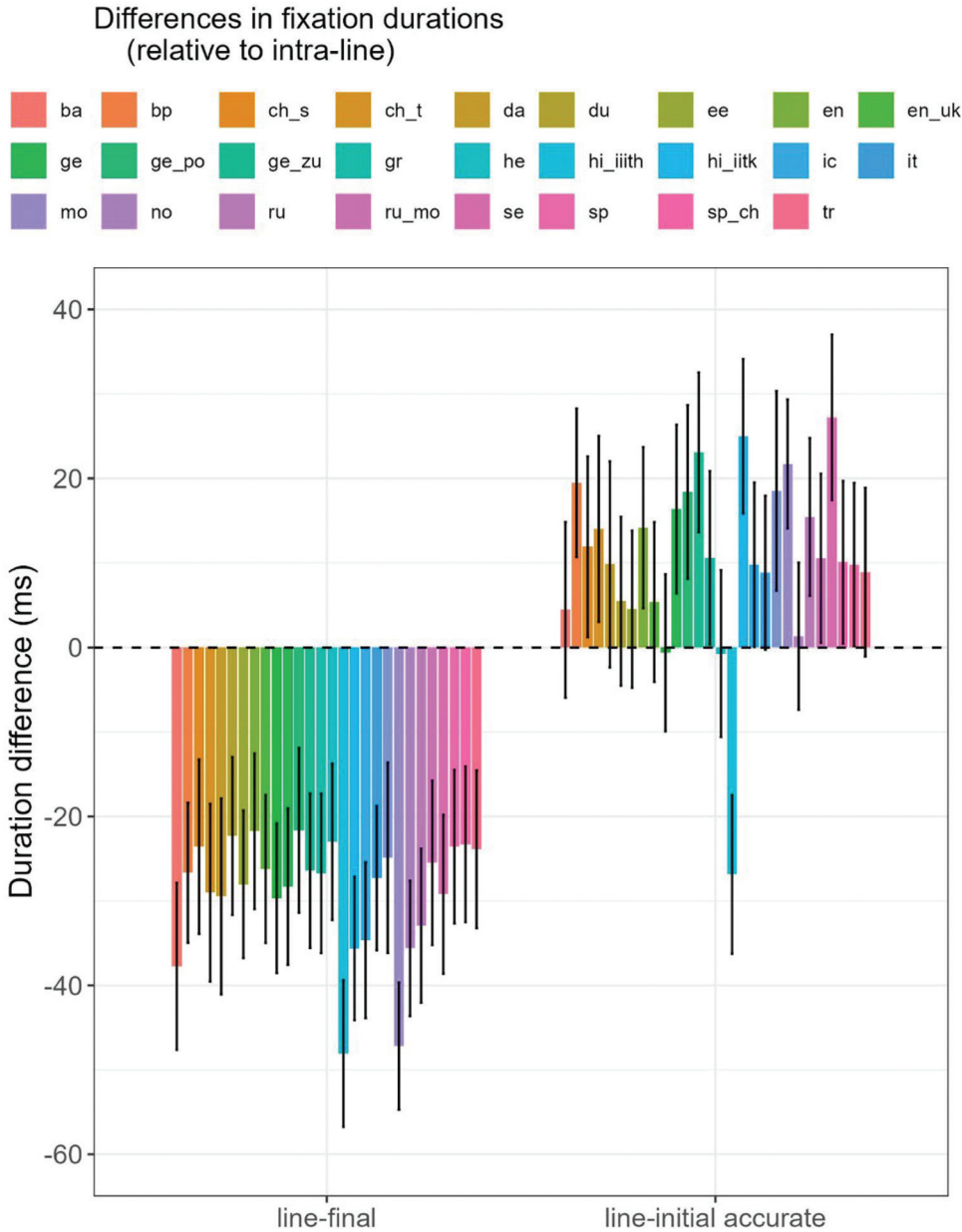
Figure 2

Partial Effects of Fixation Position Type on Fixation Duration by Language



Note. *line_initial_accurate*: line-initial accurate fixation; *line_initial_undersweep*: undersweep; *intra-line*: intra-line fixation; and *line_final*: line-final fixation. Error bars stand for ± 1 SEM. Language codes correspond to each dataset sample: ba = Basque; bp = Brazilian Portuguese; ch_s = Chinese (Mandarin, simplified); ch_t = Chinese (Mandarin, traditional); da = Danish; du = Dutch; ee = Estonian; en = English (Canada); en_uk = English (UK); fi = Finnish; ge = German; ge_po = German (Potsdam sample); ge_zu = German (Zurich sample); gr = Greek; he = Hebrew; hi_iitk = Hindi (Kanpur sample); hi_iitth = Hindi (Hyderabad sample); ic = Icelandic; it = Italian; ko = Korean; mo = Traditional Mongolian; no = Norwegian; ru = Russian; ru_mo = Russian (Moscow sample); se = Serbian; sp = Spanish; sp_ch = Spanish (Chile sample); tr = Turkish. See the online article for the color version of this figure.

Figure 3
Estimated Differences Between Durations of the Intra-Line Fixations and Line-Final Fixations (Left) and Intra-Line Fixations and Line-Initial Accurate Fixations (Right)



Note. Error bars stand for ± 1 SEM. Language codes correspond to each dataset sample: ba = Basque; bp = Brazilian Portuguese; ch_s = Chinese (Mandarin, simplified); ch_t = Chinese (Mandarin, traditional); da = Danish; du = Dutch; ee = Estonian; en = English (Canada); en_uk = English (UK); fi = Finnish; ge = German; ge_po = German (Potsdam sample); ge_zu = German (Zurich sample); gr = Greek; he = Hebrew; hi_iitk = Hindi (Kanpur sample); hi_iith = Hindi (Hyderabad sample); ic = Icelandic; it = Italian; ko = Korean; mo = Traditional Mongolian; no = Norwegian; ru = Russian; ru_mo = Russian (Moscow sample); se = Serbian; sp = Spanish; sp_ch = Spanish (Chile sample); tr = Turkish. See the online article for the color version of this figure.

types. Line-initial accurate fixations were longer than intraline fixations by an average of 12.3 ms ($SE = 0.40$, $t = -31.149$). For individual languages, line-initial accurate fixations were also numerically longer than intraline fixations in all but one case

(Hindi sample, hi_iith). This one exception is likely artifactual, because another sample of reading in Hindi (hi_iitk) that used the same materials and presentation format demonstrated patterns fully in line with the other languages. Yet the statistical

significance (at the 5% nominal level) of the difference varied by language: As Figure 3 (right) illustrates, it was reached in only a handful of languages (e.g., Brazilian Portuguese, Canadian English, and Serbian). This provides only a partial confirmation of prior reports in the literature.

In accord with the literature, last fixations on the line were significantly shorter ($p < .01$) than intraline fixations (by 30 ms on average), for both the cross-linguistically aggregated data set and for each individual language.

Undersweeps had a very restricted range of mean durations across languages (126–170 ms) and had the shortest duration on average among all positional types (139 ms, $SD = 10$ ms). Undersweeps were substantially and significantly ($p < .001$) shorter than durations of fixations in any other position (the average difference of 106 ms with accurate first fixations), both in the aggregated data and in individual languages.

To further test external validity of our findings, we compared our outcomes for the English (Canadian) sample to the data on English text reading from Parker, Tao, and Vasilev (2025). Parker et al. estimate fixation duration by position in English at 241 ($SD = 81$) ms for a line-initial fixation followed by a forward saccade, 164 (48) for an undersweep, 216 (76) for an intraline, and 186 (84) for line-final fixation. The respective numbers in the (Canadian) English data set from MECO were 235 ($SD = 88$) for a line-initial fixation followed by a forward saccade, 140 (52) for an undersweep, 221 (34) for an intraline, and 200 (91) for line-final fixation. Differences between respective positions in these two samples were not significant ($p > .1$). This finding contributes to the notion that eye movements that subserve intraline gaze transitions are qualitatively similar and replicable across studies.

Our *RQ2* concerns the direction of reading as a predictor of cross-linguistic differences in return sweeps. Inspection of durational patterns for reading in traditional Mongolian did not reveal any outstanding patterns. Although the estimated accurate first fixation duration for traditional Mongolian was one of the longest (292 ms, $SE = 3$ ms), estimated durations of fixations in all other positions for this language were long as well. As a result, the differences between fixation durations in different positions for traditional Mongolian were well within the range shown by languages with the left-to-right writing. Thus, the direction of reading did not appreciably affect the processes that go into the planning and execution of the return sweep, or the continuation of reading after the return sweep is completed.

In sum, this study confirms that changes in fixation duration as a function of fixation position in the line are stereotypical, in a set of 28 languages representing a variety of writing systems.

General Discussion

The main objective of this study was to investigate the eye movement behavior that accompanies gaze transitions between the lines of text through the cross-linguistic lens. This investigation made use of the data from 28 language samples representing 22 written languages and four writing systems—alphabet (including Hangul and traditional Mongolian with syllabic blocks), abjad, abugida, and logographic—available via the MECO project (Bao, Li, et al., 2025; Siegelman et al., 2022, 2025). This investigation was couched in the research framework that strives to determine universal and language-specific features of reading behavior (Li et al., 2022;

Liversedge et al., 2016; Siegelman et al., 2022). The cross-linguistic focus of this article stemmed from the very narrow empirical base of prior research on this topic, limited to a handful of well-researched languages like Chinese, English, and German (see the Introduction). Given the diversity of stimuli and experimental setups in our data, line breaks instantiate widely different visual and linguistic environments across the multiple language samples under consideration. Thus, the article enables an insight into whether fixation and saccadic patterns before, during, and after the gaze moves between the lines are driven by visuo-oculomotor factors, linguistic factors, or both. We hypothesized that observing relatively stable, stereotypical patterns of eye movements across diverse languages would imply the primacy of visuo-oculomotor factors and reflect the universal response of the visual system of the reader to ending a program of processing a line of text, transitioning to the next line, and initiating a new program of saccadic targeting and execution in that new line. We paid special attention to reading direction (horizontal versus vertical) as a previously uninvestigated factor of potential relevance for oculomotor control (see the Introduction for the rationale).

Specific phenomena that we considered cross-linguistically originated from the literature and were as follows (see the Introduction): (a) the first and the last fixations landing on a line only cover about 80% of the line length and neglect the visual inspection of the line-initial and line-final characters; (b) line-initial fixations are often followed by a corrective regression bringing the gaze closer to the beginning of the line; (c) line-initial fixations followed by a forward saccade are longer than intraline fixations; and (d) line-final fixations are shorter than intraline fixations. Study 1 addressed point (a), and Study 2 addressed points (b)–(d). Both studies touched upon the two research questions of the article: (*RQ1*) are phenomena (a)–(d) stereotypical and universal across languages or language-specific, and (*RQ2*) does the direction of reading (vertical in traditional Mongolian vs. horizontal in other languages) have a qualitative effect on oculomotor control around intraline transitions.

Regarding *RQ1*, our analyses both confirmed and brought some nuance to the phenomena listed as (a)–(d). All in all, patterns (a)–(d) were replicated in all languages and in the cross-linguistically aggregated data. Moreover, the numeric estimates for those patterns (the percentage of text covered between extreme fixations; the percentage of line-initial fixations followed by a corrective regression; the differences between line-initial or -final fixations relative to intraline fixations) were almost identical to the estimates reported in the prior literature solely based on English text reading (e.g., Parker & Slattery, 2025; Parker et al., 2025; Rayner, 1998). The limited variability and the high external validity of the cross-linguistic data strongly suggests that oculomotor behaviors associated with gaze transition between the lines of texts are highly stereotypical. The behaviors hold true across written languages and writing systems with drastically different visual characteristics, information content of a character, grapheme-to-phoneme correspondence, and other parameters. This constancy strongly implies that the behavioral patterns are of the visuo-oculomotor nature and reflect the response of the visual system to the line break while engaged in the task of reading for comprehension. Systematic linguistic influences on return sweeps and associated fixation patterns are highly unlikely, as they would have given rise to a substantial cross-linguistic variability which was not observed in our data.

The one correction to the received knowledge that our observations bring forward is that prior literature appears to overestimate the increase in duration of the line-initial accurate fixations compared with intraline fixations. This increase is observed numerically with only one (likely artifactual) exception, but is not statistically significant in most languages. If there is a penalty incurred on the line-initial fixation for not being ready to launch a grouped program of saccades in the direction of reading, as claimed by Rayner (1978, see quote earlier), this penalty is minimal and for most languages well within the natural variability in fixation durations.

The present cross-linguistic confirmation of the finding that roughly 10% of the words at the end of the line are not visually inspected gives rise to a question that requires further discussion. To what degree is this linguistic material processed? On one hand, the line-final characters and words are available for the parafoveal preview, which enables uptake of visual and, possibly, semantic features of upcoming words (Rayner et al., 2014; Veldre & Andrews, 2016; Yan et al., 2009). Our calculations made for alphabetic languages with Roman scripts in the Data Analysis section earlier indicate that 10% of the line length account for roughly nine characters to the right of the fixation. This expanse comfortably fits within the perceptual span of an adult proficient reader of an alphabetic language (estimated at 14–15 character spaces to the right of the fixation in English, as reviewed in Rayner, 1998). On the other hand, line-final fixations are demonstrably shorter than intraline fixations, so the visual uptake and parafoveal preview that they afford must be more limited than estimated based on prior experimentation with intraline fixations. The present corpus data are not sufficient to zoom in onto the processing details of the line-final characters and words. A proper investigation of this question may require the application of the gaze-contingent paradigm that directly manipulates the boundaries of the perceptual span. This paradigm would determine whether the entire final part of the line falls into the reader's perceptual span or there is linguistic material that is regularly omitted from visuo-perceptual processing.¹

Regarding *RQ2*, consideration of vertically read Traditional Mongolian offers a crucial test of whether reading direction shapes these mechanisms. Despite the biomechanical and neurophysiological differences between vertical and horizontal saccades, Mongolian readers exhibited spatial coverage and oculomotor patterns that were qualitatively well within the range of horizontally read scripts. This finding implies that the oculomotor system uses a shared control architecture for navigating line breaks, regardless of the direction of reading.

Taken together, the cross-linguistically uniform results of this and the previous section suggest that readers' behavior at the edges of the text lines, aimed at ensuring efficient and uninterrupted reading of visually interrupted materials, follows a highly stereotypical pattern. This stability holds true across very different writing systems and directions of reading and thus may be considered a behavioral universal. This finding aligns well with the notion that return sweeps and, broader, readers' accommodation of line transitions is chiefly visuo-oculomotor and independent of linguistic information.

¹We thank the anonymous reviewer for bringing up the point discussed in this paragraph.

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